Solution:

Mat 1302 W11 – Test III

Question	1	2	3	4	5	6	7	Total
Maximum Mark	5	2	4	2	5	4	8	30
Your Mark								

[5] Question 1. Short Answer Questions, 1 mark each. Place your answer into the boxes.

(a) Let $A = \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. Determine rank$A$. $\boxed{}$

Solution: 2, since there are two pivot columns.

(b) Let $A = \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. Dimension of the null space of A is $\boxed{}$

Solution: # of columns=rank(A)+dim(nul A). Therefore, dim(nul A)=4-2=2. You may also find a basis for nul(A) by solving $A\vec{x} = \vec{0}$ to answer this question.

(c) If $z = -6 + 8i$, then $|z| =$ $\boxed{}$

Solution: $|z| = \sqrt{(-6)^2 + 8^2} = 10$.

(d) Solve the equation $x^2 + x + 1 = 0$. Write the root(s) in $a + bi$ form.

$x =$ $\boxed{}$

Solution: $x = \dfrac{-1}{2} \pm \dfrac{\sqrt{3}}{2}i$

(e) Given that $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector of the matrix $A = \begin{bmatrix} 5 & 1 \\ 2 & 4 \end{bmatrix}$, what is the corresponding eigenvalue λ?

$\lambda =$ $\boxed{}$

Solution: Note that $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \end{bmatrix} = 6 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Therefore, $\lambda = 6$.

[2] **Question 2.** Let $\det(B) = 10$ and $\det(B^2CB^{-1}) = 20$. Find $\det(C)$.

Hint: $\det B^{-1} = (\det B)^{-1}$.

Solution: $\det(B^2CB^{-1}) = [\det(B)]^2 \det(C)/\det(B) = \det(B)\det(C) = 20$. Thus $\det(C) = 20/10 = 2$.

[4] **Question 3.** Let $z = 5 - 15i$, $w = 3 - 4i$. Compute:

1. $z + \overline{w}$

Solution: $z + \overline{w} = 5 - 15i + 3 + 4i = 8 - 11i$.

2. $\dfrac{z}{w}$. **Solution:** $\dfrac{5 - 15i}{3 - 4i} = \dfrac{(5 - 15i)(3 + 4i)}{(3 + 4i)(3 - 4i)} = \dfrac{75 - 25i}{25} = 3 - i$.

[2] Question 4. Let $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 1 \\ 2 & 0 & 1 \end{bmatrix}$. Determine the characteristic equation.

Solution: $|A - \lambda I|$

$$= \left| \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 1 \\ 2 & 0 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right| = \begin{vmatrix} 1-\lambda & 0 & 1 \\ 0 & 2-\lambda & 1 \\ 2 & 0 & 1-\lambda \end{vmatrix}$$

$$= (2-\lambda)(1-\lambda)^2 - 2(2-\lambda) = (2-\lambda)(\lambda^2 - 2\lambda - 1) = 0.$$

[5] **Question 5.** An economy consists of two sectors: Agriculture and Industry. To produce one unit of output, Agriculture requires 0.2 units of its own products and 0.3 units from Industry. Moreover, Industry requires 0.6 units from its own products and 0.4 units from Agriculture to produce one unit of output.

[1] (a) Write down the consumption matrix C.

Solution:

$$C = \begin{bmatrix} .2 & .4 \\ .3 & .6 \end{bmatrix}$$

[4] (b) Calculate the production levels required to meet a final demand of 10 units from Agriculture and 6 units from Industry.

Solution: We must solve the Leontief Input-Output Model equation

$$(I - C)\vec{x} = \vec{d}.$$

Since

$$I - C = \begin{bmatrix} 1 - .2 & -.4 \\ -.3 & 1 - .6 \end{bmatrix} = \begin{bmatrix} .8 & -.4 \\ -.3 & .4 \end{bmatrix},$$

we have

$$(I - C)^{-1} = \frac{1}{.20}\begin{bmatrix} .4 & .4 \\ .3 & .8 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ \frac{3}{2} & 4 \end{bmatrix}.$$

Therefore the production level required is given by

$$\mathbf{x} = (I - C)^{-1}\begin{bmatrix} 10 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ \frac{3}{2} & 4 \end{bmatrix}\begin{bmatrix} 10 \\ 6 \end{bmatrix} = \begin{bmatrix} 32 \\ 39 \end{bmatrix}$$

That is, Agriculture must produce 32 units and Manufacturing must produce 39 units.

[6] Question 6. Given $A = \begin{bmatrix} 1 & -3 & 2 & -4 \\ 0 & 1 & 5 & -7 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 10 & -19 \end{bmatrix}$.

[3] (a) Calculate $\det A$.

Take the first columns:

$$(-1)^{1+1}(1)\begin{vmatrix} 1 & 5 & -7 \\ 1 & 0 & 5 \\ 0 & 10 & -19 \end{vmatrix} + 0 =$$

$$(-1)^{1+1}(1)\begin{vmatrix} 0 & 5 \\ 10 & -19 \end{vmatrix} + (-1)^{2+1}(1)\begin{vmatrix} 5 & -7 \\ 10 & -19 \end{vmatrix} =$$

$$-50 - (-95+70) = -50+25 = -25$$

or use the 2nd solution.

Solution:

$$\begin{bmatrix} 1 & -3 & 2 & -4 \\ 0 & 1 & 5 & -7 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 10 & -19 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & -3 & 2 & -4 \\ 0 & 1 & 5 & -7 \\ 0 & 0 & -5 & 12 \\ 0 & 0 & 10 & -19 \end{bmatrix} \xrightarrow{R_4 + R_2} \begin{bmatrix} 1 & -3 & 2 & -4 \\ 0 & 1 & 5 & -7 \\ 0 & 0 & -5 & 12 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

Thus $\det A = 1(1)(-5)5 = -25$.

[1] (b) By your result in (a), is A invertible? Explain.

Solution: A is invertible because its determinant is nonzero.

[6] Question 7. Let $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$. Given that $|A - \lambda I| = (2-\lambda)(2-\lambda)(1-\lambda)$.

[2] (a) Find all the eigenvalues.

Solution: we have $\lambda = 2, 1, 2$.

[6] (c) Determine all eigenspaces as span sets.

Solution:

When $\lambda = 1$,

$$A - \lambda I = A - I = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \xrightarrow{R_3 - R_2, R_1 - R_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

From $(A - \lambda I)\vec{x} = \vec{0}$ it implies

$$x_2 = 0, x_1 = -x_3, x_3 \text{ is free.}$$

Hence

$$\vec{x} = x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore, $E_1 = span \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$.

When $\lambda = 2$,

$$A - 2I = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow{R_3 - R_1} \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{bmatrix} \xrightarrow{\frac{1}{-2}R_3} \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 - R_3} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

From $(A - 2I)\vec{x} = \vec{0}$ it implies

$$x_1 \text{ is free}, x_2 = 0, x_3 = 0.$$

Hence

$$\vec{x} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Therefore, $E_2 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$.

Extra page